XIAMEN UNIVERSITY MALAYSIA

Online Quiz 3

Course Code:	COM108		
Course Name:	Engineering Mathematics I		
Question Paper Setter:	Liu Mei Feng		
Academic Session:	2021/09	Question Paper:	A ☐ B ☐
Total No. of Pages:	2	Time Allocated:	2 hours
Additional Materials:	-		
Apparatus Allowed:	-		

INSTRUCTIONS TO CANDIDATES

1. There are FOUR questions in this paper. Please answer ALL questions.
2. Please follow the requirement of each question and give the corresponding answers in PDF format.
3. Please note that presentation of solutions (clarity, coherence, conciseness and completeness) is important. Show your work and organize your solution. Answer without proper justification may receive less, or even zero credit.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (5 marks)

Find the derivative of the function. [5 marks]

$$F(x) = \int_{x}^{x^3} \sinh(u^2)\, du$$

Question 2 (15 marks)

Evaluate the following integrals.

(a) [5 marks]

$$\int_{1}^{2} \frac{1}{x^2\sqrt{4-x^2}}\, dx$$

(b) [5 marks]

$$\int x\sqrt{4+x^2}\, dx$$

(c) [5 marks]

$$\int \frac{1}{x^4} e^{\frac{1}{x}}\, dx$$

Question 3 (25 marks)

Let R be the region in the first quadrant bounded by the curves $y = x^2$, $y = x + 2$, and the y-axis.

(a) Sketch the graph and indicate the region. [5 marks]

(b) Calculate the area of the region R. [5 marks]

(c) Let S be the solid obtained by rotating R about the x-axis. Find the volume of S. [7 marks]

(d) Let T be the solid obtained by rotating R about the line $x = -1$. Find the volume of T. [8 marks]

Question 4 (5 marks)

Evaluate the area under the curve by taking the limit of a Riemann sum that uses the right endpoint as the sample point. [5 marks]

$$f(x) = 2x^2 + x, \qquad [0, 2]$$

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